

Bo Hu

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EDUCATION

New York University, Tandon School of Engineering

Brooklyn, NY

May 2027 (expected)

B.S., Mathematics & Computer Science

GPA: 3.6/4.0 · Exchange: NYU Abu Dhabi (Fall 2025)

Selected Coursework: Computer Architecture, Object-Oriented Programming, Real Analysis, Ordinary Differential Equations, Waves and Optics, Electromagnetism, and Thermodynamics

Stanford University, Summer Session

Palo Alto, CA

Jun–Aug 2022

GPA: 3.8/4.0 (8 credits)

Selected Coursework: Data Structures, Astrophysics

EXPERIENCE

IT Operations Intern, Shanghai Data Exchange

Shanghai, China

Jun–Aug 2024

- Automated daily audit evidence with a cron-scheduled Selenium and Baidu OCR workflow that logs in, takes screenshots, extracts text, and autofills templates, reducing prep to 5 minutes.
- Configured Smokeping for concurrent monitoring across 40 hosts, improving coverage of packet loss and latency.
- Built an internal dashboard with Flask and MySQL to inventory TCP listening processes across servers, accelerating incident triage.

Data Analyst Intern, Neumetry Medical Inc.

Fremont, CA

Jul 2021–Aug 2022

- Supported a wearable-based cardiovascular study by onboarding participants and managing continuous physiological data collection.
- Automated Fourier-analysis scripts in Python to identify dominant signal peaks and generate plots from CSV datasets for investigators.

PROJECTS

Sysmon – Developed a lightweight Windows system monitor in *Rust*, *React*, and *Tauri* to visualize real-time CPU, memory, and disk I/O metrics; integrated IP-region lookup via Axios.

Side-Channel Attack Toolkit – Implemented a low-cost experimental setup using a high-precision ADC and instrumentation amplifier to capture power traces from AES-128 encryption on an STM32. Analyzed the traces with correlation power analysis (CPA) in *Python* to identify potential information leakage.

Finance & Economics Time-Series Analysis – Ongoing NYU Data Science Bootcamp project analyzing GDP, inflation, and interest-rate data using *Python*, *Pandas*, and *Seaborn*; performed exploratory data analysis and built correlation heatmaps to uncover macroeconomic trends.

EXTRACURRICULAR

Team Lead, NYU Robosub

Jan 2024–Present

- Motion team lead (Fall 2025–Present) of NYU’s Autonomous Underwater Vehicle team, preparing for the 2026 RoboSub competition.
- Implemented and tuned PID motion controllers with IMU and depth feedback for robust underwater trajectory tracking. Collaborated closely with the vision team to integrate *YOLO* perception models and complete mission tasks (pass the gate, torpedo deployment, path following).
- Currently extending path-planning and control pipeline to *Gazebo 11* simulation using URDF/thruster models for pre-deployment testing.

Lab Member, NYU OSIRIS Lab

Oct 2024–Present

- Member of NYU’s student-run cybersecurity research lab; lead of the SCA project, completed the OSIRIS Recruit CTF, and regularly participates in weekly meetings and CTF scrimps.

SKILLS

Languages: Python, C++, SQL, Rust, TypeScript

Frameworks: React, ROS, Flask, Tauri, Selenium

Tools: Git, Linux, Gazebo, Docker, Microcontroller

Other: Dean’s List (NYU); Bilingual — English & Mandarin